
Reconsidering technology transfer

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Abstract: 'Technology transfer' is a misnomer; this popular phrase and the studies from which it originates often suggest passivity and inertia on the part of developing country recipient firms. This paper, which uses the technological capability building system (TCB) approach as a conceptual framework, suggests that firms in developing countries exercise constrained agency in technological acquisition processes. The evidence discussed here is taken from a study of technological capability accumulation by twenty-six firms in the telecommunication sector of four developing countries: Uganda, Ghana, Tanzania and South Africa. Important insights into technological acquisition by firms in a service sector in developing countries emerged from that study, such as the existence of specific boundary management capabilities for managing relationships with suppliers; confirmation that firms use different mechanisms for acquiring codified and tacit knowledge; and support for the view that technical change and industry features have significant effects on technology acquisition strategies and their effectiveness.

Keywords: Technological capability building; technological capability accumulation; service industries; learning; developing countries; telecommunication firms; African firms; technology transfer; technology acquisition; innovation.

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Biographical notes: Gillian Marcelle has been active in the telecommunications and information and communications technology (ICT) policy arena for the past fourteen years, working in universities, telecommunication companies, regulatory authorities and as a consultant. In the last four years she has completed an extensive study exploring technology capability building processes in firms in developing countries, with an empirical focus on the African telecommunications industry. This study was carried out at Sussex University (SPRU), as part of the requirements for the award of a DPhil in Science and Technology Policy Studies, awarded in December 2002. Her previous degrees include an MBA from George Washington University and a BSc (Hons) in Economics from the University of the West Indies, St Augustine. She serves on the UN ICT Task Force, which provides advice to the UN Secretary General on the United Nations mandate and objective to use ICTs to promote sustainable human development.

1 Introduction

This paper will argue that current understanding of ‘technology transfer’ requires further specification and updating if firms in developing countries are to obtain maximum benefit from external sources of technical knowledge. The analysis presented here suggests that the processes by which firms in developing countries acquire technological capability inputs from external sources are better understood, when one takes account of the industry specific nature of technological change. This includes analysis of the endogenous factors that influence the abilities of firms in developing countries to accumulate technological capabilities through relationships with suppliers, and details the specific requirements of service sector firms. These new areas of focus form part of the technological capability building (TCB) system approach which was the conceptual framework developed in a study of technological capability accumulation by 26 firms in the telecommunication sector of four developing countries: Uganda, Ghana, Tanzania and South Africa [1].

In this approach, TCB is understood as a process of assembling or accumulating technological capabilities. It is treated as an investment activity that is not linear, sequential or orderly and that is not necessarily overly influenced by exogenous or contextual factors. Firms are the unit of analysis and the processes by which they built TCB systems, defined as a set of integrated processes and mechanisms that are used by firms to build technological capabilities over time, are centre-stage. The conceptual framework suggests that a firm’s TCB system consists of five components, namely:

- allocation of financial resources
- management practices, systems and decision making rules
- practices to establish and maintain facilitating leadership and organisational culture
- accessing external technology capability resources, from suppliers
- accessing technology capability resources from the innovation system (local and global).

Further, the approach suggests that in an *ideal-system for TCB* there is systematic and balanced operation of these five elements. A well-developed TCB system is considered to be necessary and sufficient for increasing the existing stock of technological capabilities in a firm, defined here to include both person-embodied and non-person embodied capabilities (capital equipment, software, codified knowledge systems, etc.). Firms with underdeveloped TCB systems are expected to perform poorly in capability accumulation.

This framework was applied in the empirical study of capability development by firms in developing countries, and the research themes explored included the firms’ relationships with external suppliers. The study offers insights into the technology acquisition processes of firms in developing countries, more commonly referred to as ‘technology transfer’. The rest of the paper is organised into four sections; first, there is a review of the dominant theoretical explanations of what developing countries can do to improve their performance in ‘technology transfer’. Second, there is an exposition of the process of technology acquisition suggested by the TCB system framework. Third, there is an analysis of the performance of twenty-six African telecommunication operating firms in acquiring technological inputs from external sources in the light of the TCB

system approach. The concluding section discusses the insights that emerge from this analysis.

2 Technology transfer existing concepts and evidence

2.1 Technology transfer: a misnomer

Despite its widespread usage, the term ‘technology transfer’ is problematic since it suggests passivity on the part of firms. By using the term transfer rather than acquisition, conventional researchers perhaps inadvertently impose a frame in which the role of developing countries is that of a recipient of imported technology (usually equipment) rather than that of an active economic agent searching for technological solutions. Farrell [2] and Vaitos [3] criticise the implicit assumption of passivity on the part of the recipients and purchasers of technological inputs and offer an alternative: commercialisation of technology. The TCB system framework prefers the term technology acquisition and defines this as the range of activities that are likely to be necessary for firms in developing countries to source, purchase, install, test, and commission equipment and related services from international suppliers. In the rest of this section the term ‘technology transfer’ is retained, to reflect the usage by the contributors to this field.

2.2 Deconstructing technology transfer processes

Technology transfer usually consists of commercial transactions in which a bundle of technological inputs is exchanged: equipment as well as technical services, such as technical assistance, construction, engineering and related services [4]. The prevailing view in academic literature is that firms in developing countries do not fare well in these transactions since equipment suppliers often lock developing countries into contracts of long duration and provide technical services at inflated rates. Reddy [4], reports that technological equipment (artefacts) transferred by international firms were on average four times older than those used in the home country of the transnational corporations (TNCs). It is not only developing country researchers who have provided fuel for a less than positive assessment of the results of technology transfer: research conducted from the point of view of the ‘home country’ of the TNCs has also suggested that only mature technologies should be transferred (see e.g. [5-7]).

Pavitt [6] points out that technology transfer is best understood as a process involving the cumulative flow of codified and tacit knowledge. Bell and Pavitt [8,9] take the implications of these characteristics further by suggesting that since transferred tacit knowledge is usually ‘firm-specific information concerning the characteristics and performance properties of production processes and that the recipients of product designs would normally be obliged to devote substantial resources to assimilate, adapt, and improve upon the original technology. These authors also point out that the environmental conditions facing firms in developing countries, such as poor supply conditions for resource and knowledge inputs and institutional immaturity are not conducive to sourcing external technological inputs. Studies of technological capability development in sub-Saharan Africa amply demonstrate this point for firms in this region (see e.g. 10-14)].

Technology transfer takes place in a variety of modes and at different levels within recipient firms. The main distinction is between equity-based modes of technology transfer (direct foreign investment and joint ventures) and non-equity-based modes of technology transfer (licensing, franchising, management contracts, marketing and technical service contracts, turn-key contracts and subcontracting). In the former, technology transfer takes place via intra-firm relationships and, in the latter, it is characterised as an arms-length relationship between the recipient and transferring firms. Empirical trends for the 1980s showed that direct foreign investment still accounted for the largest share of technology transfer, as measured by receipts for technology royalties, fees and technical services in the 1980s, although the context in which technology transfer takes place has changed significantly (see [15,16]). Technology transfer also takes place through trade, by importing machinery or by exporting to buyers who provide expertise [17].

Firms in developing countries are likely to face specific barriers since they often engage in technology transfer across national borders, and must respond to the need to adapt technologies to local conditions, as well as to the differences in infrastructure between home and host locations, and to distance and communication costs (see [18-20]).

2.3 Critical success factors in technology transfer

Hoffman and Girvan [21] argued that greater selectivity in policy intervention and improvements in the management of technology transfer at the national level within developing countries were possible solutions to the risks associated with the lack of genuine technology transfer. These authors also consider technology transfer to be a variegated process involving exchange of information, materials and people and explicitly recognise that the outcomes produced in technology transfer processes vary.

Studies of technology transfer suggest that firms that have been able to successfully manage technological acquisition incorporated the following elements:

- Training and learning components in technology transfer agreements and an explicit focus on acquiring various combinations of design, engineering and project management technologies [17,22].
- Sponsoring or otherwise encouraging overseas postgraduate training and work experience for engineers and managers, enabling them to acquire problem-solving skills and aptitudes, and to gain access to informal international networks [23].
- Establishing knowledge acquiring operations such as R&D centres or technological learning outposts overseas [17].
- Using management practices, corporate culture and leadership styles that facilitate efforts to jointly manage technology importation and local capability development activities [24]. Although many studies indicated that there are likely to be long-term benefits from implementing complementary approaches, empirical studies are still relatively rare [25].
- Concentrating on a wider range of activities than those associated with passive importation of technological inputs and having an awareness that passivity might lead to increased domestic production output and improvements in static efficiency but was unlikely to make a contribution to learning activities.

Bell and Pavitt [8,9] argue that it is possible for firms to develop effective systems for importing foreign technologies in combination with efforts to develop local technologies and to build technological capability. In their ideal-type, imported techniques and practices do not ‘crowd out’ local technology and domestic capability building. However, these authors noted that few countries were able to implement policy regimes that supported complementarity between domestic technological capability development and acquisition of technology from abroad; instead the majority of developing countries approached these modes of capability development as substitutes. Bell and Pavitt [8,9] present a wide range of sources of imported technology including direct foreign investment (joint ventures), subcontracting, original equipment manufacturing (OEM) agreements, licensing and contracts for know-how, designs, equipment and services. They observe that capability development may involve intensive efforts to improve and develop what is initially acquired, or more passive adaptation, or minor modifications of imported inputs.

2.4 Linearity assumptions

Lall [26,] characterises technology transfer as a process in which recipient firms through a cumulative process acquire capabilities of increasing levels of complexity. He suggests that firms move through four levels of technology transfer and acquire the associated capabilities as shown in Figure 1. This sequential approach to technology transfer forms the basis of much of the conventional academic and policy writing on this important subject and has been influential in policy development. It is complementary to the linear view of technological capability development developed by the same author.

Figure 1 Levels of technology transfer in Lall’s framework

INNOVATIVE (advanced design)	ability to develop next generation system
ADAPTIVE (technological self-reliance)	ability to adapt product design or reengineer production processes
DUPLICATIVE (intermediate)	ability to expand output without further foreign assistance
OPERATIONAL (basic level)	ability to manage/operate production facility designed and built by foreign partner

Source: (author based on [26, p.18 Table 1,27, p167 Table 1])

Stewart [28] appears to accept this linear model and uses it to offer a pessimistic view of the impact of technology transfer on development in which she argues that technology transfer between industrialised countries and developing countries is not well-suited to promoting equitable development and balanced growth in the importing country. For Stewart, since transfer involves the importation of technologies that have characteristics more suited to the home market where the technology was developed, the process can (and often does) result in the development of a dualistic society. She also argues that since only a few developing countries have been able to manage technology transfer effectively, technology transfer does not result in increased productivity, support for local technological effort and/or distribution of benefits to the majority of the population. Based on evidence from the 1980s, she also states that sub-Saharan African countries

present the most obvious example of the negative consequences of technology transfer. Stewart [28] concludes by suggesting that developing countries should adopt a staged process to improve their management of technology transfer. Borrowing from Lall [26,27], she argues that a linear model of capability development and technology transfer is useful since it defines for the developing country firm, the objective of moving phases of transfer:

“[mounting] the ladder of full assimilation...because higher stages permit more efficient operation of technology...indicates increased local technological capability, and therefore increased ability to assimilate other technologies efficiently and to acquire and bargain over new transfers, and ...to control the direction of technological change.” [28, p.309]

As will be discussed in greater detail in the next section, the TCB system approach, consistent with work by Bell and Pavitt [8]; Kim [18,19] and Ernst et al. [29], adopts a non-linear process of technology acquisition. In this approach the balance of interest between suppliers and buyers of technological inputs is assumed to be ever changing. This theoretical position is supported by evidence in Hoffman and Girvan [21], which shows that recipient countries have exercised greater degrees of freedom in managing the terms and conditions of technology transfer.

3 Technology acquisition in the TCB system approach

The TCB system framework draws on insights from existing research, and carefully considers in particular, theoretical contributions from Bell and Pavitt [8,9], Hoffman and Girvan [21], Lall [26,27] and Stewart [28]. It also draws on reviews of research on technology transfer Boseman [30], Kumar and Siddhartan [31] Radosevic [32] and Reddy [4] and empirical work reported in Kumar [15,16]. It is worth noting that much of this research is located within the manufacturing sector and does not extend to intra-firm processes. The study on which this paper is based specifically focuses on aspects of these lacunae, by examining intra-firm processes of service sector firms. This conceptual framework also builds on insights from an emerging body of work that recognises that intra-firm learning processes and the experiences of non-technology-producing users may have important lessons for understanding capability-building processes of firms in developing countries [33-36].

3.1 Technology acquisition as a non-linear process

The TCB system approach accepts that the notion of levels or stages of technology capability may be useful as an organising *rubric* to describe distinctions between kinds of capabilities, but regards the *stages* model of technology transfer as being less useful. Several reasons are advanced in support of an alternative model of technology acquisition, which relax assumptions of linearity associated with the stages model. First, by refuting the assumption that firms in developing countries have an inevitability or increasing desire to move to ever ‘higher levels of transfer’, it allows consideration of capability development objectives, other than *generating technology*. Second, it provides for a wider range of policy guidance, augmenting the recommendations that focus only on how firms in developing countries can gain access to the proprietary technologies

associated with 'higher stages' of transfer; in the non-linear TCB system approach there is a focus on policy interventions that can provide incentives for technology acquisition and increase mutual benefit to the actors in technology transactions. Third, the TCB approach focuses on what recipient firms must do to progress from operational to innovative levels of technology transfer and considers what specific factors limit or facilitate such movement at the intra-firm level. Fourth, the TCB system approach does not consider foreign partners to be, by definition, either detrimental or beneficial to the accumulation of more advanced abilities by firms in developing countries. Their role is considered instead to be an input to a process of capability development. The relationship between firms in developing countries, and foreign suppliers is considered to change over time; for example, it is assumed that the knowledge, skills and equipment provided 'with foreign assistance' can be substituted by local capabilities as firms acquire more advanced capabilities. Fifth, in this framework the nature of the technology plays a crucial role in defining the transfer relationship. In many early studies, the imbalance in capabilities between recipient and transferring firms rather than technical change was the major focus of attention. By not including technology specific factors, several important aspects of technology acquisition and the potential for strategy on the part of recipients were missed. Finally, the TCB system approach regards variation in the ability of firms to manage technology transfer as being important. When acquisition of technology takes place in a context where technological knowledge is mature, unchanging, and available from public sources and the technological frontier at which innovative activity takes place (defined as the ability to generate next generation systems) is within the grasp of firms in developing countries, some of the assumptions of linearity hold. However in more challenging contexts of technological change, alternative explanations are required. In light of these observations, the conceptual framework for understanding technology transfer relaxes most of these assumptions of linearity and is therefore more likely to be relevant in a wide range of contexts.

3.2 Integrating external and internal capability development processes

The TCB system approach recognises that, for the majority of firms in developing countries, importation of technological inputs is a major source of capability development. Internationally operating firms or local branches of such firms are the main source of supply of technological inputs and firms in developing countries use a variety of mechanisms to acquire technological capabilities from external sources. The types of mechanisms include: selecting suppliers, procuring equipment and services from external suppliers under suitable terms and conditions, and integrating this supply process with other aspects of technological capability building. Reliance on imported technological inputs is considered to be necessary because the local innovation system in the majority of developing countries does not provide adequate sources of advanced technological knowledge, equipment, software and technical services. The TCB system approach emphasises the need to balance indigenous capacity development with the importation of know-how and suggests that to effectively manage the interface with suppliers, firms must draw on many other capabilities. It is expected that firms that have developed capabilities to manage and process change, and a leadership culture that supports learning, in addition to specific management practices to manage learning will be more adept at acquiring technological inputs from external sources. In this framework, the

ability to deploy cultural leadership, and change management strategies is referred to as 'internal processes' for building technological capabilities. In the technology transfer literature these capabilities are referred to as 'absorptive' capacities or ability to 'manage technology transfer'. The TCB system approach extends the concept of absorptive capacity by specifying mechanisms used to manage boundary relationships, such as those with suppliers, and mechanisms that improve the ability and willingness to search for technological sources and the ability to integrate new varieties of technology. The framework also extends the understanding of critical success factors for acquisition of technology by focusing on the individual human aspects of technology acquisition processes. These features have been treated extensively in the organisational development literature, but have received relatively little attention in the 'technology transfer' literature [37].

3.3 *Constrained agency arising from technical change and industry features*

The effectiveness of the acquisition of technologies by firms in developing countries is considered explicitly to be under the partial control of these firms. Other important factors, such as the willingness and ability of external suppliers to play a role in capability development and the nature of the technological inputs that the firm in the developing country is seeking to acquire will also matter. Therefore, the technology acquisition process is considered to be a boundary relationship in which firms in developing countries exercise *constrained agency*. The firms are not completely passive actors at the mercy of TNCs, but also are not fully in control of the extent to which they can maximise capability development objectives using externally acquired inputs. This characterisation has features in common with the treatment provided by Bell and Pavitt [8,9]. The amplification provided by this approach is that it provides a detailed analysis of the conditions that influence the acquisition of different types of capabilities, and it emphasises industry specific factors as one of the explanatory factors.

The limitations and opportunities for making effective use of supplier relationships for technological capability building are expected to be derived from the nature of the specific technological inputs being sought and the willingness and ability of supplier firms to provide these inputs. In the conventional technology transfer field represented here by Lall [26,27]; and Stewart [28] firms in developing countries are often assumed to have less access to technological inputs as a result of the concentration of innovation activity in the industrialised countries. The secular trend of the increasing concentration of innovation activity is not questioned here, but an alternative reading of its consequences for firms in developing countries is offered. It is argued that while the inputs for generating radical innovation may be affected by this trend, many of the capabilities that are required by firms in developing countries are likely to be unaffected by concentration in innovative activities. Therefore, the majority of firms in developing countries that do not operate at technological frontiers are unlikely to be constrained by the increasing concentration of innovative activity. Many of the early studies of technology transfer focused on the manufacturing sector, where firms in developing countries were seeking to produce the same output (at lower cost) as their suppliers. The trends for radical innovation and incremental innovative activity to become increasingly concentrated in large, transnational corporations that invest heavily in professional R&D facilities and other supporting infrastructures, and that seek to appropriate returns through the enforcement of intellectual property rights would have an impact on the

manufacturing sector. This threat of the extension of proprietary rights to production technologies may have been overstated. In service sectors, the purchaser of imported inputs often does *not* compete directly with its suppliers, and the threat of non-disclosure of technological functionality etc. is therefore reduced [39].

This approach takes account of the trends identified by Bell and Pavitt [8,9] such as the increasing specialisation in many technological markets and notes that this has led to an increase in the sources of technological inputs, particularly for processes other than radical innovation. For instance, there are many more suppliers of codified knowledge and there have been increases in the number of specialised suppliers of equipment. Other factors that may facilitate access to external inputs include increased international migration of skilled labour and information and communication technologies that assist in the technological search process. The TCB system approach also considers that technological change has implications for the relationship between suppliers of technological inputs and their customers. In the telecommunication industry, a specific technological change that has major implications is the increases in the embeddedness of knowledge within equipment.

By presenting a more detailed analysis of the types of technological capabilities that are likely to be desired by firms in developing countries, and by considering the constituent elements of technological capabilities, the TCB system approach provides a more nuanced analysis of the technological acquisition process. Specifically, the approach considers that international suppliers can provide only a subset of the capability inputs required from external sources. Suppliers operating in commercial markets are likely to be able to supply tacit and codified knowledge, software, equipment, etc., but cannot provide the supplier management capability that firms require to integrate these inputs. A fully operational capability to manage supplier relationships requires active investment and components that cannot be provided by suppliers. Effectiveness in managing supplier relationships includes the ability to search for alternative sources, to negotiate supply on appropriate terms and conditions and to integrate external inputs from a variety of sources. Another key feature of the TCB system approach to understanding the role of technological acquisition in capability building is the emphasis on understanding how the changing nature of the relationship between supplier and user and the technological characteristics of the inputs affect the process of technological acquisition. For example, sector specific studies on capability development in the telecommunication sector provide useful insights into how the acquisitions of capability inputs from external sources is impacted by industry specific factors and technological change [40-43]. Conventional studies of technology transfer often do not adequately take these factors into account.

3.4 Evidence from African telecommunication firms

This section analyses the evidence provided by the empirical study of technological capability development in African telecommunication firms. It begins with a summary of the specific mechanisms used by sample firms and then examines the empirical evidence from sample firms in the light of themes that are suggested by the TCB system approach.

3.5 *Supplier selection*

Operating companies in the sample considered cost effectiveness, product functionality, depth of technological knowledge, technical support, track record in similar markets and speed of delivery to be among the most important criteria. The sample firms with well developed TCB systems expressed agency by having defined criteria, and supplier selection systems that included technology assessment and evaluation mechanisms, whereas firms with underdeveloped TCB systems did not. There was a reasonably good fit between the criteria reportedly used by operating companies and the perception of their suppliers. The analysis of empirical results also confirms that operating firms did not consider the ability to generate products with a high degree of technological novelty as an important criterion, but placed more emphasis on commercial abilities and the execution abilities of their suppliers. Where there were technical criteria, these were in terms of the functionality of the equipment that would be supplied. Embeddedness of knowledge and information in equipment was one of the features of technological change in the telecommunication industry that influenced the acquisition of capability inputs from suppliers of equipment. The companies reportedly used as suppliers are shown in Table 1.

Table 1 Identifying important suppliers of equipment and services

<i>Company Name</i>	<i>Number of operating companies</i>
Ecrisson	4
Alcatel, Cisco, Lucent, NEC, Scientific Atlanta	2
Advent, Airspan UK Ltd (formerly DSC), AT&T, BBC BT, Digital Equipment Corporation, Divicom, ITELCO, MAS, MCL, Mitsubishi, MSI, NDS, Nortel, NTL, Plessey, Siemens, Tadiran, Varian Fujitsu, Hughes Network Systems, Iredeto, Iridium	1

N=16 operating companies

3.6 *Specific mechanisms for acquiring technology from suppliers*

Eleven firms provided data on the specific mechanisms used to acquire information, know-how and skills from their suppliers of equipment and services; these data are presented in Table 2. Analysis of these data suggests that formation of joint network design teams with suppliers and tendering and bid evaluation processes were used by the highest number of sample firms for technology acquisition. Exclusive product demonstrations, long term attachments at the suppliers' site and interaction through social networks were among the mechanisms that were each used by one firm. There were variations among firms according to size and level of development of TCB system. For example one large national telecommunication company reported a very different pattern, insofar as the company placed its emphasis on transfer during field trials and procurement, rather than via provision of regular training programmes.

Table 2 Specific routines used for technology transfer from suppliers

<i>Technology transfer mechanism</i>	<i>Frequency (#of firms reporting usage)</i>
1 formation of joint project teams for network planning	4
2 tendering and bid evaluation processes including development of technical specifications	4
3 intensive use of supplier technical support hotline	3
4 regular communication at functional middle level management with international and local divisions of supplier firms	3
5 short term contracts for expatriate engineers from supplier companies during testing & commissioning on site in African country	2
6 training on specific equipment provided at supplier premises in South Africa (REGIONAL CENTRE)	2
7 training on specific equipment provided at supplier premises in Sweden	2
8 technical support services provided by supplier staff on operating company site	2
9 general technological training courses organised by suppliers and delivered overseas	2
10 know-how transfer projects managed as part of procurement and equipment trial processes	2
11 designation of executive with specific responsibility for managing supplier relationship	2
12 long term attachments with supplier companies at their premises	1
13 shared social networks	1
14 exclusive product demonstrations	1

N= eleven operating companies

3.7 Specific and complementary routines for managing supplier relationships

Firms are not *born* with the ability to manage supplier relationships. The development of *boundary management competences* required active investment and implementation of routines to strengthen:

- the ability to *evaluate* technological capability requirements
- the ability to *scan and search*; which incorporates the ability to investigate and understand technological trends, and the ability to select high quality sources of technological inputs on appropriate terms and conditions
- the ability to *integrate* external inputs

Only seven firms out of twenty-six, used formal evaluation and/or scan and search routines. These seven firms had well-developed TCB systems, consisting of a wide range of TCB mechanisms and also paid attention to the integration of all capability development activity. South African operating companies were the most intensive users

of formal technology search and evaluation techniques with five out of the firms operating in South Africa reporting the use of these mechanisms. The relatively large size of the South African companies may be an explanatory factor here, since scan and search activities require a critical mass of highly skilled technical personnel. The other distinctive feature of the large South African operating companies is that their network development was funded independent of bilateral or multilateral development assistance. Whereas in Uganda and Tanzania, the large public network operators did not operate independent search and scan mechanisms. It may be, as reported by the interviewees, that these firms were not able to exercise choice in selection of suppliers, but were obliged to use the suppliers named by the financiers. This characteristic was also the case for the Ghanaian public operator prior to privatisation.

In addition to the South African companies, there were two companies in the sample that reported using formal search and evaluation techniques. These two firms were *outliers* in their active approach to capability development. The Ugandan company that used formal search and evaluation routines had a technology and operational strategy that was closely influenced and directed by its major shareholder; a large South African company, with operations in other parts of the continent. In the case of the other outlier, a small Tanzanian data-communication company, the implementation of formal search and evaluation was linked directly to the corporate culture of the firm. This company was part of a group of companies, founded by a group of young Africans, including MIT engineering graduates. The background of the founders, their interest in technological sophistication, vision in promoting innovation and their direct involvement in influencing technological strategy had enabled this small company to extend its boundaries beyond the limits imposed by its small size and the deficiencies of the local innovation system.

If informal scan and search routines, such as attending trade fairs and exhibitions and participation in different communities of interest (professional bodies, trade associations, working parties of regional and international organisations), are included, the number of firms increases to eighteen. The pattern for use of informal mechanisms differed from that for formal mechanisms in so far as there was wider representation from Ugandan, Ghanaian and Tanzanian firms. For example, a small Tanzanian private network operating company reported that its owner used trade fairs and exhibitions as a cost-effective mechanism for acquiring technological information. The owner of the firm reported that he took direct and personal responsibility for this activity drawing on his technological background. This individual was a technological pioneer in Tanzania and had a background in aviation and aeronautical engineering and he used this foundation to expand into telecommunication. Another example comes from a small Ghanaian data communication company, whose spokesperson reported an exceptionally high usage of these informal mechanisms. This firm was an active participant in technological development at the regional and international levels. The background of the founder of this business was as a professional engineer with more than 20 years private sector overseas experience in the USA, and active involvement with United Nations development agencies as a technical assistance provider. This may account for the permeability of this firm's organisational boundaries and its high propensity to be actively engaged in technology scanning and search activities. The other intensive users of these informal mechanisms were large public network national operators in Uganda and Tanzania and the national mobile operators in South Africa. However, the explanations for the observed patterns differ for these two groups. For the former, active participation in industry associations and in the working parties of the African

Telecommunication Union, Commonwealth Telecommunication Organisation and the International Telecommunication Union, was reported to be an important means for independent assessment of technological trends and evaluation of suppliers. These regional and multilateral bodies were considered by interviewees to be 'honest brokers' since they did not fund network development, but instead were mandated to share information about appropriate technological choice and to provide information on equipment standardisation. Conversely, for the two national mobile operators in South Africa, participation in industry associations was said to be focused on groupings such as the GSM Association, where employees engaged in the development of standards for particular technologies on an equal footing with the other participants. The relatively strong standing of the South African participants is demonstrated by the fact that employees of one of the operators had held international office in the GSM Association at the time the data was collected.

Another community of interest, which appears to have been an important source of external inputs, is the strategic investors and shareholders in the operating companies. Even companies with only moderate development of their TCB system and those with no discernible TCB system, reported that they maintained regular communication with regional headquarters, sister companies and shareholders as a means of gaining external technological input. There were some large companies, particularly in Ghana that used formal mechanisms for acquiring technological inputs and expertise from shareholders and strategic investors, including long-term attachments for local staff at the site of the investors and formal training courses organised by the investor companies.

In summary, the evidence drawn from interviewees shows that the sample firms engaged in activities that were designed to develop their scan and search capabilities. The foregoing analysis suggests that variations in the use and implementation of these mechanisms are related to features such as size, organisational culture and leadership. The factors that appear to have facilitated firms in integrating technological inputs from external sources are considered next.

The TCB system approach suggests that the ability to integrate externally sourced technological inputs is likely to require *complementary* capabilities if firms are to benefit from boundary relationships [44,45]. As noted in the earlier discussion of the conceptual framework, these features tend to be discussed in organisational development theory, but ignored by economists examining technical change and innovation. The empirical evidence supported this proposition; sample firms were found to be ineffective at supplier management when complementary aspects of technological evaluation and integration were not in place. In particular, firms that created conditions for experimentation, reinforcement of knowledge and promotion of higher order learning were better able to integrate external inputs. Among the sample firms, there were firms with explicit technology strategies, which considered themselves to be world-class as well as the technologically active outlier firms, as discussed earlier in this section. There was also a large Ghanaian firm that had focused its competitive strategy on the development of the technological competences of its people. This company focused its evaluation efforts on formal needs assessments. Similarly, firms that demonstrated a high level of competence in integrating external inputs were also those that exhibited advanced competencies in management development and practices such as clear definition and assignment of responsibilities.

3.8 Role of human factors in accessing tacit knowledge

Evidence from the sample firms confirmed that supplier relationships were an important source of technological capabilities. In particular, these contractual relationships were reported to be used to acquire equipment, software, and *codified* knowledge and information in the form of equipment handbooks, training course materials, maintenance procedures and protocols. Although the equipment component typically accounted for a large percentage of the reported financial cost, contracts usually included elements of non-embodied capability. While the majority of sample firms used traditional delivery mechanisms for accessing codified knowledge and information, a few firms augmented these with information technology tools such as hotlines, and web-based technology support. The traditional and computer assisted mechanisms seem to have provided operating companies with access to factual information on equipment functionality and specifications, maintenance routines and equipment and software upgrades.

Embodied capabilities were provided when suppliers organised training courses and provided technical support and services. A variety of mechanisms were used for the transfer of embodied capabilities and these ranged from formal, classroom based instruction at the operating company site to long-term attachments of staff of operating companies at the site of the supplier. The other mechanisms for embodied capability acquisition from suppliers reported to be used by the sample firms included site visits to supplier premises and reference sites, establishing joint project teams for network development, and use of consultants and specialists from suppliers on short term technical assistance assignments. The most significant characteristic of these mechanisms for acquiring embodied capabilities was that they involved the exchange of *tacit* knowledge and information between supplier and operating company. Because of the importance of the tacit component of knowledge flow, these exchanges took place using both formal and informal mechanisms and their effectiveness of exchange appears to have been highly dependent on design and implementation. Tacit knowledge and information exchanges require trust between parties, acknowledgement of the contextual characteristic of knowledge and information, and shared meanings. As a result, the design of the exchange mechanisms that was sensitive to these characteristics would require considerable skill on the part of *both* the operating companies and the supplier firms.

The features of technology acquisition exchanges that appeared to improve effectiveness, particularly for transfer of tacit knowledge components, included mechanisms for:

- selecting training instructors and ‘experts’ who had up-to-date knowledge sets, sound fundamental technological training, good communication skills, and were experienced in similar operating contexts
- ensuring that there is regularity of contact over all phases of network development and operation and not limiting the tacit knowledge and information exchanges to commissioning and testing phases
- organising for joint ownership of the technology acquisition objectives expressed in the design and staffing of project teams and the breadth of the activities that are assigned to buyers of technological inputs.

Other characteristics of the management of supplier relationships that reportedly contributed to the effectiveness in the sample firms' TCB efforts included:

- the ability of personnel in the telecommunication operating companies to maintain strong social networks with the staff of supplier firms
- the maintaining of regular contact between staff of operating companies and the supplier firms at all levels of hierarchy, ranging from global executives and local managers
- implementation of mechanisms that foregrounded joint learning such as through equipment trials, where suppliers and operating companies were actively involved in understanding technology requirements, and specifications prior to full commercial deployment
- implementation of tendering processes that resulted in maximum disclosure of codified information and provided opportunities for intensive communication between suppliers and users.

When there were weak search and scan abilities, the operating companies in the sample were less well able to select suppliers that were skilled in the design of technology dissemination projects. The involuntary lack of control over choice of suppliers on the part of the operating company had the same effect as weak scan and search capabilities by artificially restricting suppliers. This lack of control over design of technology acquisition projects appears to have limited the effectiveness of exchange of codified and tacit information flows and, particularly, hampered the flow of tacit information. Another important design flaw was caused by the misconception that technology acquisition was limited to commissioning, testing and installation of equipment. Smaller firms in the sample reported that they were hampered by their lack of critical mass of qualified and experienced personnel and their inability to make appropriate selections to staff technology acquisition projects.

3.9 Impact of technological change

The sample firms' reliance on global market leaders as an important source of external technological inputs appeared to be related to the following features of technological change in the telecommunication industry. As more knowledge and information is encoded or embedded in equipment and software control algorithms, suppliers appeared to have been forced to become more expert in helping their customers to understand and use this sophisticated equipment. While this characteristic of embeddedness has the effect of making knowledge required for network management increasingly product (equipment/application) specific, it also increases the possibility for fine-tuning network performance characteristics through software changes. This increasing embeddedness of knowledge in equipment has meant that the supply of embodied and non-embodied elements of technological capabilities in the telecommunication industry has become increasingly coupled. This trend has had the effect of reducing the importance of intermediaries and information brokers as sources of external technological inputs. Telecommunication operating companies expect their supplier to design and support equipment with appropriate functionalities and to understand the operational context in which that equipment will be used.

There has also been a rapid and accelerating pace of technological change in telecommunication systems subsystems and sub-components that has led to the need for regular, continuous interface between suppliers and operating companies. The pace of change has also had some negative consequences in that it has created artificial crises spurring operating companies into continuous rounds of technological upgrading and 'improvement'.

Finally, as a result the high levels of software controls used in the switching and access networks and the high information and knowledge content coded in equipment (devices and components), IT and project management have become core competences for telecommunication network operating companies. All professional technical employees and particularly those in specific functions such as network management and optimisation are required to have high levels of IT literacy.

In summary, the effect of technological change in the telecommunication industry has led firms in developing countries, such as those in the sample, to be more likely to use a single supplier firm as a source of external technological inputs. The single source of supply is likely to allow operating companies to take advantage of the coupling effect of having access to non-embodied capabilities and embodied capabilities. These long-term, one-stop supplier relationships were common among the sample firms but did not necessarily lead to the deleterious effect on access to technological capabilities that was implied by conventional studies on technology transfer. There appeared to be potential shortcomings in the single-supplier mode, especially for firms without technological evaluation capabilities, insofar as the operating companies could be persuaded to make regular upgrades in technological inputs at a pace determined by their supplier, rather than at a more measured pace in line with their capacity to direct and absorb the integration of these inputs. Technological change also led to the emergence of at least two areas of core competence: IT and project management.

3.10 Impact of specialisation and concentration of innovative activity

The findings confirm that the sample firms, including those with extensive search routines, made most frequent use of global market leaders as their suppliers (these firms were those that had led the trend to concentrate innovative activity as measured by R&D, patents, etc.) in the equipment industry. The suppliers used by the sample firms mainly included market leaders and second-tier equipment suppliers. The evidence does not provide support for a view that firms in developing countries were active in diversifying their suppliers. The qualitative accounts suggested that the operating companies in the sample did not favour smaller or alternative sources of supply. Where small companies such as Tadiran, the Israeli based company, were used, these firms were often niche market leaders in specific segments, such as wireless in the local loop access network components and systems. This pattern is consistent with the nature of competition in the telecommunication equipment supply industry.

Put simply, the global market leaders were willing and able to provide the operating companies with access to up-to-date equipment and the technical services required to efficiently develop, deploy and maintain networks based on that equipment. Therefore, these global market leaders were the suppliers of choice of the operating companies. The evidence from the sample firms does not support the view that implementing search routines leads to greater diversification. Ironically, for the operating firms where there were multiple sources of technological inputs, this was involuntary and was imposed by

multilateral agencies and heavily criticised by the purchasers. The qualitative accounts, however, do support the argument that the firms that had developed routines for technology needs assessment and integration of external inputs were better able to manage their single-supplier relationships.

In summary, in this study the telecommunication operating companies in four developing countries selected a limited range of companies to supply their key technological inputs. While the operating companies appeared not to have an interest per se in the innovative performance of the supplier companies, their keen interest in functionality and standardisation of equipment led to a de facto limit on the range of supplier sources. This is the same outcome as if they had been explicitly interested in choosing only those companies with relatively strong innovation performance. The limited range of suppliers does not appear to have affected the ability of these firms in developing countries to exercise 'constrained agency' in acquiring technological inputs from these firms. This may be explained by the interest of the global market leaders in increasing their effectiveness as suppliers of non-embodied and embodied capabilities.

The suppliers to the sample firms appeared to perceive that their success in winning business from the telecommunication operating companies in Africa was dependent on their *responsiveness* to the business needs of these companies. The perception that it was important to have joint ownership of technological development objectives between suppliers and operating companies was indicated by the inclusion of such features as the ability to jointly plan and manage network deployment and operation among the list of critical success factors reported by suppliers.

What is perhaps even more interesting is the evidence that the supplier firms were themselves investing in organisational systems that may improve their ability to be responsive to the technological development objectives of their developing country customers. For example, one of the supplier firms provided evidence that it had developed specialised career paths for technical assistance experts who were deployed to assist customers with network deployment and management. This company had also developed mechanisms whereby the employees of their customers could undertake formal certified training courses to achieve comparable levels of skill and expertise as the career network specialists of the supplier firm. There is also corroborating evidence that suggests that this company considered its ability to design these technology dissemination mechanisms as a source of core competence in all market segments, and had applied these business processes in developed markets [46]. The evidence from this research suggests that this company was extending implementation of these business processes to its developing country markets. The organisational innovations that were perceived by supplier firms in the sample to be most effective include:

- undertaking investments in their internal technological learning and facilitation of knowledge dissemination
- adaptation of business processes to increase knowledge and information flows between suppliers and users
- regular and continuous interfaces with customers through a variety of formal and informal mechanisms
- documentation of the best practice mechanisms for knowledge dissemination
- improved account management
- use of computer assisted tools for dissemination of information.

In summary, the changing nature of competition appears to be closely associated with the technological features of the products offered by these firms. Competitive success was considered to be increasingly dependent on the ability to assist customers in achieving their objectives and was believed to require organisational innovation to improve and maintain levels of responsiveness. On the evidence, it would appear that the global market leaders were satisfying their customers in developing countries in terms of their levels of responsiveness. This result suggests an interesting line of further research that would investigate whether relationships similar to the ones exhibited in the telecommunication industry are present in other sectors.

3.11 Limits on suppliers as source of technological inputs

Although equipment and service suppliers appear to have been an effective and significant source of technological inputs for the operating firms in the sample, there are indications of some types of knowledge inputs where these private sector companies were less useful. Private sector equipment and service companies provided non-embodied and embodied technological inputs that were required by operating companies, but were regarded as being less effective as sources of knowledge about fundamental scientific principles, basic technological training and understanding about the nature and direction of technological trends. The operating companies also had needs for this type of knowledge and information that could not be satisfied through their relationships with suppliers. This 'know-why' was probably more effectively sourced from innovation system institutions such as universities, training colleges and research laboratories as well as from policy and regulatory bodies. The empirical evidence also suggests that international and regional bodies might play a useful role in providing this information and knowledge. Other communities of interest and practice also made an important contribution to supply this 'know-why'.

Private-sector suppliers also did not seem to be particularly effective in providing access to knowledge and information about the best sources of codified and tacit knowledge needed by operating companies, also referred to as 'know-who.' The communities of interest and practice referred to earlier appeared to be particularly good at providing independent assessments of competing sources of technological inputs, as were specialist information providers and organisers of trade fairs and exhibitions. To make use of these sources of 'know-who', operating companies had to have routines for evaluation and for searching and scanning. When operating companies were using additional sources of external capability inputs, the equipment suppliers would be partners in technological choice, rather than taking direct ownership of this critical function. Private sector equipment suppliers would be unlikely to be an effective source of 'know-who' because they would be unlikely to provide objective assessments of the range of sources available. These results have policy implications for the design and functioning of national innovation systems.

4 Concluding remarks

Several key insights emerged from the study. First, the capability to *manage* supplier relationships was found to be an important factor that influences whether firms in developing countries are able to benefit from access to external sources of technological

capabilities. Effective management of supplier relationships was found to be strongly influenced by endogenous variables such as the level of development of the firm's technological capability building system. In particular, the extent to which firms had acquired the specific competences of technological evaluation, search, acquisition, and integration appeared to influence strongly the ability of firms to benefit from commercial relationships with external suppliers. Complementary competences that relied on human factors were found to be important for effective boundary relationship management; a component that is too often ignored in economics based studies of 'technology transfer.'

Second, firms in developing countries appeared to exercise *constrained agency* in their management of these relationships. The sample firms were similar to the majority of firms in developing countries in that they were not generators of radical technological innovation at the frontiers of technology, but many were required to use, operate and adapt state-of-the-art technological inputs in their production processes. This was found to require sophisticated user capabilities, some of which can be sourced through relationships with suppliers, and others that require internal development. In addition, it was found to be important to differentiate between the many different types of knowledge and to assess the potential value that supplier relationships can add in each case. These considerations are particularly relevant in the case of service sector firms in developing countries, where many components of technological capability are embedded in equipment.

An important conclusion of the study was that exogenous factors such as technological change, increasing specialisation, concentration of innovation and the nature of competition also influence the contribution of supplier relationships to technological capability building objectives of firms in developing countries. On balance, the evidence suggests an optimistic interpretation of the effects of these exogenous factors. It provides illustrations of how large and small firms in developing countries can interact with global market leaders to the benefit of their technological objectives. The confluence of technological and market dynamics appears to be leading customers and suppliers to become keenly interested in the exchange of knowledge and this seems to be leading to more open dissemination and greater disclosure.

The empirical study outlined here confirms insights from research within the development studies field that suggest that external sources of technological capability need not crowd out domestic capability development. It extends this observation by highlighting the conditions under which the acquisition of technological capability inputs can advance the technological capability development objectives of firms in developing countries.

These results have important implications for policy recommendations for firms in developing countries, national science and technology institutions and international organisations that support capability development. The analysis indicates that firms can improve technological capability accumulation by paying attention to supplier management as a *specific* aspect of capability development. There is also room for improvement in terms of the management of interaction with the national systems of innovation, particularly to complement knowledge acquisition from commercial sources.

Public-sector bodies in the national innovation system can play an important role in the capability building process by supplying complementary types of knowledge, e.g. 'know-why' and 'know-who' to support the search and scan capability of firms. To become more effective in this role, the national systems of innovation in developing

countries will require considerable *strengthening* in terms of the range of institutions involved and the tools used to support capability development in firms.

These policy recommendations for firms in developing countries and the public sector bodies in the national innovation systems can contribute to the transformation of the capability development effort, and to more effective investments in learning.

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